

TrES-3b

R-0369

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_230624 · created 2026-07-18T16:04:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460119.9158 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.018
Transit depth (Rp/Rs)^2	3.8 +/- 0.71 [%]
Orbital Inclination (inc)	81.58 +/- 0.53
Airmass coefficient 1 (a1)	1.003 +/- 0.01
Airmass coefficient 2 (a2)	-0.0023 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#2 - [534, 408]
Optimal Aperture	4.06
Optimal Annulus	12.55
Transit Duration (day)	0.0555 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.018 vs 0.16309 (deviation 1.77σ)
Duration fit vs published	0.0555 d vs 0.0567 d (deviation 0.2σ)
Mid-time O–C	-0.3 min
Depth SNR	10.8
Residual scatter	1.02 %

Light curve

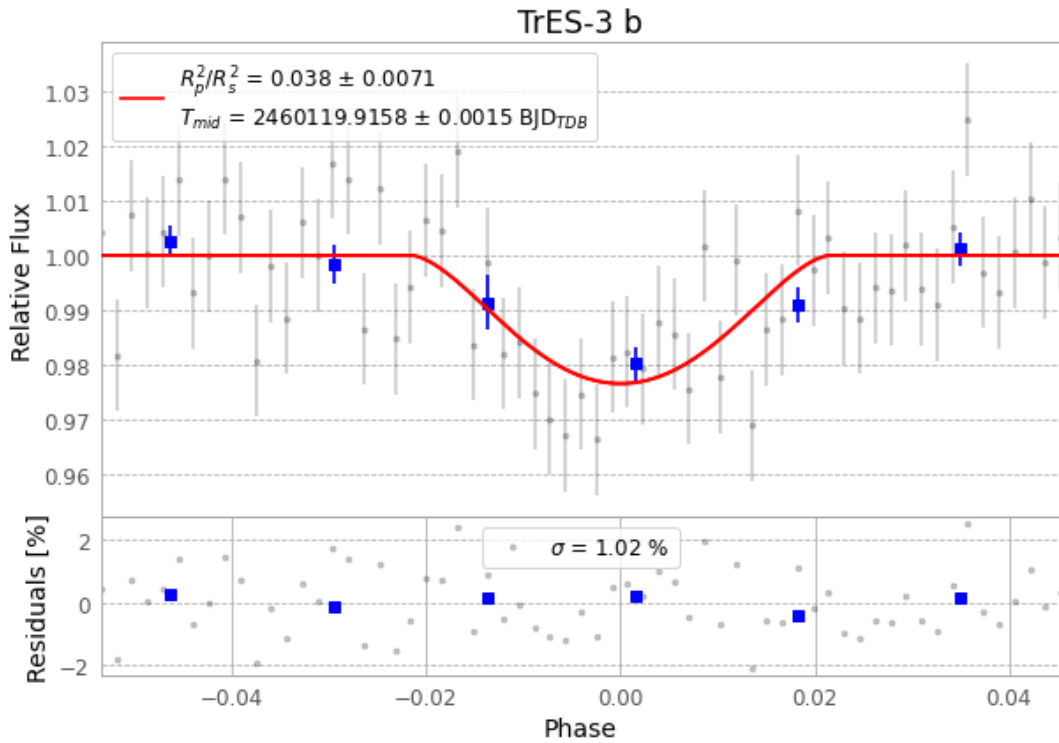


Figure 1 — FinalLightCurve_TrES-3 b_2023-06-24.png (SHA-256 63f36ae933495434...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 35ed6aff25e7f343...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

63 FITS frames (41 MB), TRES-3230624081215.FITS → TRES-3230624111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230624.log (f69997d0a781...), FinalLightCurve_TrES-3 b_2023-06-24.png (63f36ae93349...), FinalLightCurve_TrES-3 b_2023-06-24.csv (8fcc8caa42a0...)

Compute resources

Wall time 115.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 16:04Z.